

A PROJECT TITLE

ON

Air Quality Index Prediction Using Ridge and Lasso Regression
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Name of Title: Air Quality Index Prediction Using Ridge and Lasso Regression

1. Abstract:

Air pollution is one of the most significant environmental challenges faced worldwide, affecting human health, ecosystems, and overall quality of life. The increasing concentration of harmful pollutants in the atmosphere has made the monitoring and prediction of air quality an essential task for governments and environmental organisations. The Air Quality Index (AQI) is a widely used indicator that measures the level of air pollution and its potential impact on public health. Accurate prediction of AQI enables timely preventive measures and supports informed decision-making for environmental management. This project, titled "Air Quality Index Prediction Using Ridge and Lasso Regression," focuses on developing a machine learning-based model for predicting AQI values using Python.

The proposed system utilises Ridge Regression and Lasso Regression, two regularised linear regression techniques that enhance prediction accuracy while addressing issues such as overfitting and multicollinearity datasets. The model is trained using air quality parameters, including particulate matter (PM_{2.5} and PM₁₀), sulphur dioxide (SO₂), nitrogen dioxide (NO₂), carbon monoxide (CO), ozone (O₃), temperature, humidity, and other relevant environmental factors. Data preprocessing techniques such as handling missing values, feature scaling, and data normalisation are employed to improve model performance and reliability. Ridge Regression minimises prediction errors by applying L₂ regularisation, whereas Lasso Regression employs L₁ regularisation to perform feature selection by reducing the influence of less significant variables. The predictive performance of both models is evaluated using standard metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared (R²) score. Comparative analysis is conducted to determine the most effective approach for AQI prediction. The proposed system provides a practical and efficient solution for forecasting air quality conditions, contributing to environmental monitoring, public health awareness, and sustainable urban planning through the application of machine learning techniques.

2. INTRODUCTION:

Air pollution has become a major environmental issue that affects human health and the ecosystem worldwide. The increasing levels of harmful pollutants in the atmosphere have made air quality monitoring and prediction an important area of research. The Air Quality Index (AQI) is a standard measure used to indicate the level of air pollution and its potential impact on public health. Accurate prediction of AQI can help individuals, organisations, and governments take timely preventive measures to reduce the adverse effects of air pollution. Machine Learning techniques have significantly improved the ability to analyse environmental data and predict air quality with high accuracy. Among various regression methods, Ridge Regression and Lasso Regression are effective techniques for predicting continuous values while reducing problems such as overfitting and multicollinearity. These models provide reliable predictions and improve the overall performance of the system. This project, titled "**Air Quality Index Prediction Using Ridge and Lasso Regression,**" is developed using Python to predict AQI values based on environmental parameters such as PM_{2.5}, PM₁₀, CO, NO₂, SO₂, and O₃. The proposed system performs data preprocessing, model training, and performance evaluation using standard metrics. The project aims to provide an efficient and practical solution for AQI prediction, contributing to better environmental monitoring and public health awareness.

SOFTWARE REQUIREMENTS:

The software requirements for the development and implementation of the project titled "Air Quality Index Prediction Using Ridge and Lasso Regression" are listed below:

- Operating System: Windows 10/11 or Ubuntu 22.04 (or above)
 - Programming Language: Python 3.10 or above
 - Development Environment: Visual Studio Code (VS Code) or Jupyter Notebook
 - Machine Learning Libraries: Scikit-learn (for implementing Ridge and Lasso Regression models)
 - Data Processing Libraries: Pandas and NumPy (for data cleaning, preprocessing, and analysis)
 - Data Visualization Libraries: Matplotlib and Seaborn (for graphical representation of data and model performance)
 - Dataset Format: Comma-Separated Values (CSV) files
 - Package Manager: pip (for installing the required Python packages)
 - Version Control Tool (Optional): Git and GitHub
 - Web Browser: Google Chrome or Microsoft Edge (for viewing outputs and documentation)
- Python Packages Used
- Pandas
 - NumPy
 - Scikit-learn
 - Matplotlib
 - Seaborn
 - SciPy (optional, for statistical computations)

HARDWARE REQUIREMENTS

The hardware requirements for the implementation of the project "Air Quality Index Prediction Using Ridge and Lasso Regression" are as follows:

- Processor: Intel Core i3 (8th Generation) / AMD Ryzen 3 or above
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 20 GB of free disk space
- Operating System: Windows 10/11 or Ubuntu 22.04 and above
- Display: 1366 × 768 resolution or higher
- Internet Connection: Required for downloading datasets, Python libraries, and project dependencies
- Keyboard and Mouse: Standard input devices for development and testing

These hardware specifications are sufficient for running Python, performing data preprocessing, training Ridge and Lasso Regression models, and visualizing the prediction results efficiently.

Software Installation Requirements :

The required Python libraries can be installed using the following command:

```
pip install pandas numpy scikit-learn matplotlib seaborn SciPy
```

These software tools and libraries provide an efficient environment for data preprocessing, machine learning model development, performance evaluation, and visualization of Air Quality Index predictions. They ensure accurate implementation of Ridge and Lasso Regression techniques while supporting reliable and scalable project development in Python.

Course Instructor